

Multiple Choice (10 marks)

1. The belts and zones of Jupiter are
 - (a) names for the layers of gaseous and metallic hydrogen deep within the planet.
 - (b) alternating bands of rising and falling air at different latitudes.
 - (c) alternating regions of charged particles in Jupiter's magnetic field.
 - (d) regions of the plasma torus created by ions from Io's volcanoes
2. What is the significance of the 1:2:4 resonance in the Jupiter's moons system?
 - (a) The resonance pulls Io in different directions and generates heat.
 - (b) It makes the orbit of Io slightly elliptical.
 - (c) It creates a gap with no asteroids between the orbits.
 - (d) It prevents formation of the ring material into other moons.
3. The two moons of Mars are called ...
 - (a) Tritos and Desmos
 - (b) Tritos and Deimos
 - (c) Phobos and Tritos
 - (d) Phobos and Deimos
4. How are planetary rings made?
 - (a) From the dismantling of small moons by impacts
 - (b) From fragments of planets ejected by impacts
 - (c) From dust grains that escape from passing comets
 - (d) From accretion within the solar nebula at the same time the planets formed
5. Why doesn't Venus have seasons like Mars and Earth do?
 - (a) Its rotation axis is nearly perpendicular to the plane of the Solar System.
 - (b) It does not have an ozone layer.
 - (c) It does not rotate fast enough.
 - (d) It is too close to the Sun.

True / False (10 marks)

Answer True or False

6. True or False: When the Moon is setting at noon, it is in the third quarter phase, positioned opposite the Sun in the sky.
7. True or False: The axis of the Earth is tilted at an angle of approximately 23.5 degrees relative to its orbital plane around the Sun.
8. True or False: Organisms living deep in the oceans around seafloor volcanic vents and in hot springs most resemble the common ancestor of all life.
9. True or False: Venus exhibits evidence of surface processes including impacts, erosion, and volcanism.
10. True or False: A Mars year is approximately 2 Earth years long, reflecting its longer orbital period around the Sun.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of Hawking radiation and explain how it contributes to the gradual evaporation of black holes over time, including its implications for our understanding of black hole thermodynamics.
12. Discuss the significance of Betelgeuse within the constellation Orion, focusing on its characteristics, cultural importance, and role in astronomical studies.

End of Paper